

Projected Heat Kernels from MTT Fixed-Point Data

Conditional Sector Identities and the Physical-Replacement Boundary

Peter Nero

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Abstract

This paper isolates the exact projected-heat-kernel statement available from declared fixed-point data. On a specified compact Riemannian internal space, Euclidean problem, or spatial Cauchy-slice Hilbert space, let A be a nonnegative self-adjoint elliptic operator, let P be a reducing orthogonal projector, and let $\tau > 0$. Functional calculus then uniquely defines

$$K_{\text{MTT}}(x, y; \tau) = \langle x | P e^{-\tau A} P | y \rangle.$$

We prove smoothness in the elliptic spectral setting, covariance under unitary equivalence, coherent-band error estimates, and the joint distributional limit to the full identity. The harmonic case is important: if $AP = 0$, then $P e^{-\tau A} P = P$, so τ creates no additional smoothing on the retained sector. Gap damping concerns $e^{-\tau A} Q$; the projected operator $P e^{-\tau A} P$ instead annihilates Q .

These facts do not authorize a universal replacement of spacetime deltas, canonical commutators, gauge constraints, or local vertices. Such a replacement is a new physical model unless a selected realization supplies the operator domain, A , P , and τ , and separately proves locality or causal support, covariance, gauge/BRST compatibility, and the relevant algebraic identities. The result is therefore a conditional execution interface, not a derivation of a universal MTT regulator.

Version 1 Revision Note

Supersedes The unversioned manuscript *Canonical Coherent Kernels from MTT Fixed-Point Data*.

Reason The earlier manuscript did not declare whether A acted internally, on a spatial slice, or in Euclidean signature; it blurred harmonic projection with heat smoothing and promoted a sector kernel to a universal delta, CCR, source, and interaction replacement.

Resolution This version types the operator domain, proves explicitly that $AP = 0$ gives $P e^{-\tau A} P = P$, separates Q -sector damping from the projected kernel, and adds locality, covariance, causal, gauge, and algebra-preservation gates to every physical use.

Retained result Functional calculus uniquely determines the projected heat operator and displayed spectral error bounds from compatible realization data.

Remaining boundary Current MTT fixed-point theory does not select one such triple for every physical sector or prove that replacing a standard distribution preserves the target theory's causal and gauge structure.

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1 Purpose and claim discipline

The earlier delta-projection papers established statements of the form

$$K_\epsilon \longrightarrow \delta$$

or

$$\text{finite object} \longrightarrow \text{sharp standard object}.$$

Those statements explain why delta distributions recur across physics, but they do not yet select a physical finite kernel. The missing execution-level direction is

$$\text{MTT fixed-point data} \implies K_{\text{coh}} \implies \text{finite correction}.$$

This paper proves that direction in the fixed-point regime.

Conditional construction. After a realization has specified the operator domain and compatible data (A, P, τ) , the projected heat operator is not chosen freely. Whether that operator represents a physical source, identity, or regulator is a separate theorem for the target sector.

What is proved

We prove, in a standard compact spectral setting and in the corresponding Hilbert-space operator language, that fixed-point data (A, P, τ) determine a canonical bounded/smoothing operator

$$B_\tau := P e^{-\tau A} P$$

and hence a kernel K_{MTT} . We give explicit error estimates comparing B_τ with the full identity and with the coherent-sector identity.

What is not claimed

We do not claim that all numerical values of τ , spectral gaps, or carrier-induced operators have been computed, or that an internal stabilization operator automatically acts on external spacetime variables. Once a fixed-point sector supplies a typed (A, P, τ) , the corresponding projected heat kernel is fixed within that realization. No statement about an unrelated delta distribution follows without an intertwiner identifying the two domains.

2 Analytic setting

Let X be a compact smooth Riemannian manifold without boundary, or a compact manifold with boundary equipped with a self-adjoint elliptic boundary condition. More generally, one may replace scalar functions by sections of a Hermitian vector bundle; for readability we use the scalar notation.

Let

$$A \geq 0$$

be a nonnegative self-adjoint Laplace-type operator on $L^2(X)$ with compact resolvent. Let

$$A\phi_n = \lambda_n\phi_n, \quad 0 \leq \lambda_0 \leq \lambda_1 \leq \cdots,$$

be an orthonormal spectral resolution, with eigenvalues repeated according to multiplicity.

Assumption 2.1 (Fixed-point spectral data). We are given:

- (i) a bounded orthogonal projector P on $L^2(X)$;
- (ii) $Q := I - P$;
- (iii) commutation $PA = AP$ on $\text{Dom}(A)$;
- (iv) an incoherent-sector gap

$$A|_{\text{Ran } Q} \geq \lambda^* Q \quad \text{for some } \lambda^* > 0;$$

- (v) a proper-time/coherence scale $\tau > 0$.

Remark 2.2. In the fixed-point papers P is the coherent projector Π_{coh} , often defined by Riesz spectral calculus on the fiber/internal operator. The gap λ^* is the uniform positive spectral gap above the coherent modes. The operator A is the linearized damping or elliptic operator controlling stabilization near the fixed-point sector.

Remark 2.3 (Domain typing). The compact elliptic theorem is Riemannian. It may describe an internal fiber, a Euclidean problem, or an operator on a spatial Cauchy slice after those data are supplied. It is not by itself a Lorentzian spacetime heat kernel. Passing to a hyperbolic or retarded problem requires a separate domain, support, and wavefront-set analysis.

Definition 2.4 (Canonical MTT coherent operator). Under [theorem 2.1](#), define

$$B_\tau := P e^{-\tau A} P.$$

When B_τ has an integral kernel, we write

$$K_{\text{MTT}}(x, y; \tau) := \langle x | B_\tau | y \rangle.$$

Because P commutes with A , the spectral form is transparent. If $P\phi_n = p_n\phi_n$ with $p_n \in \{0, 1\}$, then

$$B_\tau f = \sum_n p_n e^{-\tau \lambda_n} \langle \phi_n, f \rangle \phi_n,$$

and the kernel is

$$K_{\text{MTT}}(x, y; \tau) = \sum_{n: p_n=1} e^{-\tau \lambda_n} \phi_n(x) \phi_n^*(y).$$

3 Existence, smoothness, and covariance of the canonical kernel

Theorem 3.1 (Kernel existence and smoothing). Assume [theorem 2.1](#). For every $\tau > 0$, $e^{-\tau A}$ is a smoothing trace-class operator. Consequently $B_\tau = P e^{-\tau A} P$ is bounded and trace class. If P is a spectral projector of A , then B_τ has a smooth kernel

$$K_{\text{MTT}}(\cdot, \cdot; \tau) \in C^\infty(X \times X).$$

If P is finite rank, the same conclusion holds even at $\tau = 0$, with $B_0 = P$.

Proof. Since A is a nonnegative self-adjoint elliptic operator with compact resolvent, the spectral theorem gives

$$e^{-\tau A} f = \sum_n e^{-\tau \lambda_n} \langle \phi_n, f \rangle \phi_n.$$

For every $N \geq 0$,

$$A^N e^{-\tau A}$$

is bounded, because $\lambda^N e^{-\tau \lambda}$ is bounded on $[0, \infty)$. Elliptic regularity then implies that $e^{-\tau A}$ maps distributions to smooth functions. On a compact manifold, $e^{-\tau A}$ is trace class for $\tau > 0$, and has a smooth heat kernel. Multiplication on both sides by the bounded spectral projector P , which commutes with A , preserves trace class and smoothness of the kernel. If P is finite rank, P is a finite sum of smooth eigenfunction projectors, hence has a smooth kernel at $\tau = 0$. $\square$

Theorem 3.2 (Re-encoding covariance). Let $U : L^2(X) \rightarrow L^2(X')$ be a unitary re-encoding. Define

$$A' = U A U^{-1}, \quad P' = U P U^{-1}.$$

Then the canonical coherent operator transforms as

$$B'_\tau = P' e^{-\tau A'} P' = U B_\tau U^{-1}.$$

Thus the construction of B_τ depends only on the fixed-point operator data up to unitary re-encoding.

Proof. Functional calculus is covariant under unitary conjugation:

$$e^{-\tau A'} = e^{-\tau UAU^{-1}} = Ue^{-\tau A}U^{-1}.$$

Therefore

$$P'e^{-\tau A'}P' = (UPU^{-1})(Ue^{-\tau A}U^{-1})(UPU^{-1}) = U(Pe^{-\tau A}P)U^{-1}.$$

□

4 Sector identity and error estimates

The canonical kernel should not be confused with the full Dirac identity kernel. It is an identity-like object on the retained coherent sector, with explicit error terms.

Assumption 4.1 (Coherent bandwidth). For some $\Lambda_{\text{coh}} \geq 0$,

$$\sigma(A|_{\text{Ran } P}) \subset [0, \Lambda_{\text{coh}}].$$

Remark 4.2. If P is the harmonic zero-mode projector, then $\Lambda_{\text{coh}} = 0$ and $B_\tau = P$ exactly. If P retains a finite coherent band, then Λ_{coh} is the largest retained coherent eigenvalue.

Proposition 4.3 (Harmonic-projector degeneracy). *If $AP = PA = 0$, then*

$$Pe^{-\tau A}P = P$$

for every $\tau \geq 0$. Thus the proper-time parameter produces no further smoothing inside the harmonic sector.

Proof. Functional calculus gives $e^{-\tau A}P = P$ when $AP = 0$, and left multiplication by P leaves P unchanged. □

Theorem 4.4 (Coherent-sector identity estimate). *Assume [theorems 2.1](#) and [4.1](#). For every $f \in L^2(X)$,*

$$\|Pf - B_\tau f\|_{L^2} \leq (1 - e^{-\tau \Lambda_{\text{coh}}}) \|Pf\|_{L^2}.$$

Consequently,

$$\|f - B_\tau f\|_{L^2} \leq \|Qf\|_{L^2} + (1 - e^{-\tau \Lambda_{\text{coh}}}) \|Pf\|_{L^2}.$$

Proof. Since P commutes with A , $B_\tau f = e^{-\tau A}Pf$. Thus

$$Pf - B_\tau f = (I - e^{-\tau A})Pf.$$

On $\text{Ran } P$, the spectral values of A lie in $[0, \Lambda_{\text{coh}}]$. Hence

$$\|(I - e^{-\tau A})P\|_{L^2 \rightarrow L^2} = \sup_{\lambda \in \sigma(A|_{\text{Ran } P})} |1 - e^{-\tau \lambda}| \leq 1 - e^{-\tau \Lambda_{\text{coh}}}.$$

This proves the first estimate. The second follows from

$$f - B_\tau f = Qf + (Pf - B_\tau f)$$

and the triangle inequality. □

Corollary 4.5 (Small- τ sector correction). *Under [theorems 2.1](#) and [4.1](#),*

$$\|Pf - B_\tau f\|_{L^2} \leq \tau \Lambda_{\text{coh}} \|Pf\|_{L^2}.$$

Proof. Use $1 - e^{-x} \leq x$ for $x \geq 0$. □

Theorem 4.6 (Incoherent damping estimate). *Assume [theorem 2.1](#). Then*

$$\|e^{-\tau A} Q\|_{L^2 \rightarrow L^2} \leq e^{-\tau \lambda^*}.$$

More generally, for $s \geq 0$,

$$\|A^{s/2} e^{-\tau A} Q\|_{L^2 \rightarrow L^2} \leq \sup_{\lambda \geq \lambda^*} \lambda^{s/2} e^{-\tau \lambda}.$$

Proof. On $\text{Ran } Q$, the spectral theorem and the gap assumption give $\lambda \geq \lambda^*$. Therefore

$$\|e^{-\tau A} Q\| = \sup_{\lambda \in \sigma(A|_{\text{Ran } Q})} e^{-\tau \lambda} \leq e^{-\tau \lambda^*}.$$

The higher estimate is identical after multiplying the spectral multiplier by $\lambda^{s/2}$. □

5 Recovery of the Dirac delta

The canonical MTT kernel is finite. The Dirac delta is recovered only under an idealizing limit.

Let $P_N := \mathbf{1}_{[0, \Lambda_N]}(A)$ be an increasing family of spectral projectors with $\Lambda_N \rightarrow \infty$. Let

$$B_{N, \tau} := P_N e^{-\tau A} P_N.$$

Theorem 5.1 (Joint sharp limit). *Let $f \in C^\infty(X)$. Suppose $\Lambda_N \rightarrow \infty$ and $\tau_N \downarrow 0$. Then*

$$B_{N, \tau_N} f \rightarrow f$$

in $C^\infty(X)$. Equivalently, the kernels of B_{N, τ_N} converge to $\delta(x - y)$ in the distributional sense on $X \times X$.

Proof. Write

$$f - B_{N, \tau_N} f = (I - P_N) f + P_N (I - e^{-\tau_N A}) P_N f.$$

The first term tends to zero in all Sobolev norms because spectral projectors converge strongly to the identity on smooth functions, and smooth spectral coefficients decay rapidly. For the second term, fix a Sobolev index s . Choose $M > s$. Since $f \in C^\infty$,

$$\sum_n (1 + \lambda_n)^M |\langle \phi_n, f \rangle|^2 < \infty.$$

For each fixed n , $1 - e^{-\tau_N \lambda_n} \rightarrow 0$. The summand is dominated by a multiple of $(1 + \lambda_n)^M |\langle \phi_n, f \rangle|^2$ for N large enough and M chosen above s . Dominated convergence gives convergence to zero in H^s . Since s is arbitrary, Sobolev embedding gives convergence in C^∞ . The distributional kernel statement is the usual kernel form of convergence to the identity. □

Corollary 5.2 (Meaning of the delta limit). *The Dirac delta is not the fixed-point kernel itself. It is the joint idealization in which:*

- (i) *coherent bandwidth becomes complete;*
- (ii) *proper-time width tends to zero;*
- (iii) *the incoherent complement is no longer discarded.*

At finite bandwidth the declared model object is K_{MTT} , not the full identity distribution. Calling it physically selected requires a source theorem for the particular realization.

6 Eligibility contract for downstream use

The mathematical construction gives an execution rule only after the target distribution has been identified with the same typed sector.

Definition 6.1 (Eligible projected-kernel substitution). A downstream identity or source distribution is eligible for substitution only when: (i) it acts on the same Hilbert or bundle domain as A and P ; (ii) a selected intertwiner identifies the target variables with that domain; and (iii) the substitution preserves every required symmetry, constraint, support, and algebraic identity. Under those hypotheses the declared model may use

$$\delta(x - y) \rightsquigarrow K_{\text{MTT}}(x, y; \tau) = \langle x | P e^{-\tau A} P | y \rangle.$$

Within the heat/proper-time class this is not an arbitrary smoothing prescription. The eligibility hypotheses, however, are additional physical content and cannot be inferred from functional calculus alone.

Theorem 6.2 (Uniqueness within the heat/proper-time class). *Among positive self-adjoint contraction semigroups generated by the fixed-point operator A , the one-parameter family $e^{-\tau A}$ is uniquely determined by A . Therefore, once P and τ are fixed, the operator $B_\tau = P e^{-\tau A} P$ is unique.*

Proof. By the spectral theorem, a nonnegative self-adjoint operator A determines a unique strongly continuous contraction semigroup $T_\tau = e^{-\tau A}$. Conversely, the generator of such a semigroup is unique by the Hille–Yosida theorem. Therefore the proper-time filter associated with the fixed-point generator is fixed. Multiplying on the left and right by the fixed coherent projector P gives a unique B_τ . $\square$

Remark 6.3. The theorem does not say that no other regulator can be written down. It says that if the kernel is required to be the heat/proper-time filter generated by the same operator governing MTT stabilization, then the kernel is fixed by the fixed-point data.

7 Finite corrections in standard structures

7.1 Green functions

Let L be a positive self-adjoint invertible operator commuting with A and P . The ordinary point-source Green equation

$$LG = I$$

has kernel form

$$LG(x, y) = \delta(x - y).$$

The MTT coherent Green operator is

$$G_\tau^{\text{MTT}} := L^{-1} B_\tau.$$

It satisfies

$$LG_\tau^{\text{MTT}} = B_\tau,$$

or in kernels,

$$L_x G_\tau^{\text{MTT}}(x, y) = K_{\text{MTT}}(x, y; \tau).$$

Proposition 7.1 (Green-function correction). *Assume L^{-1} is bounded on the Hilbert space under consideration. Then*

$$\left\| L^{-1} - G_{\tau}^{\text{MTT}} \right\| \leq \left\| L^{-1} \right\| \left\| I - B_{\tau} \right\|.$$

On a coherent-band input $f = Pf$,

$$\left\| L^{-1} f - G_{\tau}^{\text{MTT}} f \right\| \leq \left\| L^{-1} \right\| (1 - e^{-\tau \Lambda_{\text{coh}}}) \|f\|.$$

Proof. The first estimate follows from

$$L^{-1} - L^{-1} B_{\tau} = L^{-1} (I - B_{\tau}).$$

The second estimate uses [theorem 4.4](#). □

7.2 Canonical commutators

The sharp canonical commutator

$$[\phi(x), \pi(y)] = i\hbar \delta(x - y)$$

belongs to the symplectic algebra of the field theory. Writing instead

$$[\phi_{\text{coh}}(x), \pi_{\text{coh}}(y)] = i\hbar K_{\text{MTT}}(x, y; \tau).$$

defines a projected or modified symplectic algebra; it is not a consequence of the heat-kernel theorem. A physical use must prove nondegeneracy on the retained quotient, covariance, microcausality, and compatibility with the field equations. Without those checks the formula is only a candidate finite-sector model.

7.3 Contact vertices

A local n -point contact vertex can be written schematically as

$$V_{\delta}^{(n)}(x_1, \dots, x_n) = \int_X \prod_{j=1}^n \delta(x - x_j) \, dx.$$

A finite-overlap candidate is

$$V_{\text{MTT}}^{(n)}(x_1, \dots, x_n; \tau) = \int_X \prod_{j=1}^n K_{\text{MTT}}(x, x_j; \tau) \, dx.$$

At nonzero width this is generally a nonlocal effective interaction. It requires a local parent mediator or an independent EFT, gauge/BRST, and causality analysis before it can replace a local vertex.

7.4 Measurement effects

The distributional position effect

$$|x\rangle\langle x|$$

suggests a positive finite candidate effect with kernel

$$E_x^{\text{MTT}}(y, z; \tau) = K_{\text{MTT}}(y, x; \tau) K_{\text{MTT}}^*(z, x; \tau).$$

To obtain a measurement model, these effects must form a normalized POVM and must be coupled to a declared apparatus instrument. The projected kernel alone does not supply either requirement.

7.5 Noise kernels

If an independently selected temporal operator A_t and memory scale τ are supplied, they define the candidate correlation kernel

$$C_{\text{MTT}}(t, s) = \langle t | e^{-\tau A_t} | s \rangle$$

rather than an exact white-noise delta. In Markov limits this may approach

$$D\delta(t - s),$$

but neither A_t nor its identification with the internal fixed-point operator follows from the elliptic theorem above.

8 Worked example: circle with massive operator

Let $X = S^1$ with coordinate $\theta \in [0, 2\pi)$, and take

$$A_m = -\partial_\theta^2 + m^2, \quad m > 0.$$

The eigenfunctions are

$$\phi_n(\theta) = \frac{1}{\sqrt{2\pi}} e^{in\theta}, \quad \lambda_n = n^2 + m^2.$$

Let

$$P_N = \sum_{|n| \leq N} |\phi_n\rangle \langle \phi_n|.$$

Then

$$K_{N,\tau}^{\text{MTT}}(\theta, \theta') = \frac{1}{2\pi} \sum_{|n| \leq N} e^{-\tau(n^2 + m^2)} e^{in(\theta - \theta')}.$$

For fixed N , this is a smooth finite kernel. For $N \rightarrow \infty$, it becomes the heat kernel of A_m :

$$K_{\infty,\tau}^{\text{MTT}}(\theta, \theta') = e^{-\tau m^2} \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-\tau n^2} e^{in(\theta - \theta')}.$$

As $\tau \downarrow 0$, this tends distributionally to the periodic delta

$$\delta_{S^1}(\theta - \theta').$$

The finite correction to a coherent-band identity is explicit:

$$\|P_N f - B_{N,\tau} f\|_{L^2}^2 = \sum_{|n| \leq N} \left(1 - e^{-\tau(n^2 + m^2)}\right)^2 |f_n|^2.$$

Thus the correction is mode-resolved and directly computable from the fixed-point spectral data.

9 Programmatic consequence

The previous delta papers established the forward structural dictionary:

$$\text{finite kernel} \longrightarrow \delta.$$

This paper supplies the fixed-point reverse direction:

$$(A, P, \tau, \lambda^*) \longrightarrow K_{\text{MTT}} \longrightarrow \text{finite correction}.$$

Execution principle. For an eligible same-domain occurrence, use the projected heat/proper-time kernel supplied by the declared realization and retain the target theory's symmetry, support, and algebra checks as explicit obligations.

This makes the delta dictionary computational inside a declared sector. Point sources, commutators, contact interactions, measurement effects, and noise kernels are not thereby identified with one another; each requires its own admissible intertwiner and consistency theorem.

10 Conclusion

The central mathematical question is whether declared fixed-point data determine a specific projected heat kernel. They do:

$$B_\tau = P e^{-\tau A} P$$

and hence

$$K_{\text{MTT}}(x, y; \tau) = \langle x | P e^{-\tau A} P | y \rangle.$$

This kernel is smooth in the compact elliptic setting, covariant under unitary equivalence, and identity-like on a retained finite band with explicit error bounds. In the harmonic case it reduces exactly to P . The separate semigroup $e^{-\tau A} Q$ is gap-damped. A family with complete bandwidth and vanishing proper time tends to the full identity distribution.

Thus the program is no longer merely:

$$\delta = \text{singular limit of some finite object}.$$

The rigorous conclusion is therefore:

$$\boxed{\text{typed fixed-point data determine a conditional sector kernel.}}$$

Promoting that kernel to a physical modification remains a realization-specific source and consistency problem.

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